

C805

Pathophysiology

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 10-year-old who plays in a youth orchestra develops extreme thirst, frequent urination, and weight loss over several weeks. Tests show very low insulin and antibodies against pancreatic islet cells. Her grandfather has type 2 diabetes. What is the underlying mechanism?

- A. Insulin resistance in muscle and fat cells.
- B. Overproduction of glucagon by alpha cells.
- C. Too much insulin production.
- D. Autoimmune destruction of pancreatic beta cells.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A carpenter who builds custom cabinets gets a splinter, and within hours the area is red, warm, swollen, and painful. He is right-handed. What mainly causes the swelling?

- A. Vasoconstriction in the injured area.
- B. Bacterial toxins directly dissolving the tissue.
- C. Antibody production by B cells within hours.
- D. Increased vascular permeability that lets fluid and proteins leak into the tissue.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A 68-year-old retired mail carrier with long-standing high blood pressure has crackles in both lungs and shortness of breath when lying flat. His ankles are not swollen. He enjoys gardening. Which is most likely?

- A. Pneumonia.
- B. Left-sided heart failure.
- C. Right-sided heart failure.
- D. No heart failure, since heart failure always causes ankle swelling.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Which finding points to right-sided heart failure?

- A.** Jugular vein distention and swelling in the legs.
- B.** Pink, frothy sputum.
- C.** Crackles and shortness of breath when lying flat.
- D.** Chest pain spreading to the jaw.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A teacher at an elementary school with a peanut-free cafeteria is stung by a bee and within minutes has hives, wheezing, and low blood pressure. Which hypersensitivity type is this?

- A.** Type II (cytotoxic).
- B.** Type III (immune complex).
- C.** Type IV (delayed, T-cell).
- D.** Type I (IgE-mediated).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A rash appears 48 hours after skin contact with poison ivy. Which mechanism is responsible?

- A.** A direct chemical burn from the plant.
- B.** Type IV, T-cell mediated hypersensitivity.
- C.** Type III, immune complexes.
- D.** Type I, IgE and mast cells.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A 55-year-old warehouse manager who smokes has crushing chest pain that has lasted 40 minutes and is not relieved by rest. His favorite hobby is fishing. What is the most likely underlying process?

- A.** Plaque narrowing alone, without any clot.
- B.** Pericarditis.
- C.** A ruptured coronary plaque with a clot blocking blood flow, causing heart muscle death.
- D.** Stable angina from brief ischemia.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A 70-year-old former factory worker with a 40-pack-year smoking history has a barrel chest and trouble breathing out. He lives with two cats. Spirometry shows a reduced FEV1/FVC ratio. What is the main problem in emphysema?

- A. Stiff lungs that cannot expand.
- B. Excess mucus from enlarged bronchial glands.
- C. Destruction of alveolar walls with loss of elastic recoil, trapping air.
- D. An allergic reaction to the cats.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

After surgery at a community hospital with a new MRI scanner, a patient who received large doses of opioids is breathing slowly. ABG: pH 7.30, PaCO₂ 55 mmHg, HCO₃ 26 mEq/L. The surgery lasted 2 hours. Which imbalance is present?

- A. Respiratory acidosis.
- B. Metabolic alkalosis, because HCO₃ is 26.
- C. Metabolic acidosis.
- D. Respiratory alkalosis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

ABG: pH 7.48, PaCO₂ 40 mmHg, HCO₃ 32 mEq/L. Which imbalance is present?

- A. Normal, because PaCO₂ is normal.
- B. Metabolic alkalosis.
- C. Metabolic acidosis.
- D. Respiratory alkalosis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

In a long-time smoker who sings in a choir, the normal ciliated columnar cells lining the bronchi are replaced by stratified squamous cells. He also walks every day. What adaptation is this?

- A.** Metaplasia.
- B.** Dysplasia.
- C.** Hyperplasia.
- D.** Hypertrophy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

A weightlifter's skeletal muscle cells grow larger without increasing in number. Which adaptation is this?

- A.** Metaplasia.
- B.** Atrophy.
- C.** Hypertrophy.
- D.** Hyperplasia.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A 10-year-old ~~who plays in a youth orchestra~~ develops extreme thirst, frequent urination, and weight loss over several weeks. Tests show very low insulin and antibodies against pancreatic islet cells. ~~Her grandfather has type 2 diabetes.~~ What is the underlying mechanism?

Target word: mechanism **Decides:** Islet autoantibodies and very low insulin.

A	Camouflage	Describes the grandfather's type 2, not her findings.
B	Adjacent category	Not the primary defect in type 1 diabetes.
C	Inverted operation	Her insulin is very low.
D	Correct	Type 1 diabetes is autoimmune loss of insulin-producing cells.

2. Answer D Concept 2

A carpenter ~~who builds custom cabinets~~ gets a splinter, and within hours the area is red, warm, swollen, and painful. ~~He is right handed.~~ What mainly causes the swelling?

Target word: cause of swelling **Decides:** Early inflammatory vascular changes.

A	Inverted operation	Inflammation causes vasodilation after a brief constriction.
B	Adjacent category	Swelling comes from the body's vascular response.
C	Confusion set	Adaptive antibody responses take days.
D	Correct	Leaky vessels produce edema.

3. Answer B Concept 3

A 68-year-old ~~retired mail carrier~~ with long-standing high blood pressure has crackles in both lungs and shortness of breath when lying flat. His ankles are not swollen. ~~He enjoys gardening.~~ Which is most likely?

Target word: likely condition **Decides:** Lung congestion signs.

A	Adjacent category	Can cause crackles, but orthopnea with long-standing hypertension points to the heart.
B	Correct	Blood backs up into the lungs, causing crackles and orthopnea.
C	Inverted operation	Would cause body signs such as JVD and ankle edema.
D	Absolute language	Left-sided failure can occur without edema.

4. Answer A Concept 3 • transfer

Which finding points to right-sided heart failure?

Target word: reverse: pick the right-sided sign **Decides:** Systemic venous backup.

A	Correct	Right-sided failure backs blood into the body's veins.
B	Confusion set	A sign of pulmonary edema from left-sided failure.
C	Inverted operation	Left-sided signs.
D	Adjacent category	Suggests ischemia, not right-sided failure.

5. Answer D Concept 4

A teacher ~~at an elementary school with a peanut free cafeteria~~ is stung by a bee and within minutes has hives, wheezing, and low blood pressure. Which hypersensitivity type is this?

Target word: hypersensitivity type **Decides:** Rapid systemic reaction within minutes.

A	Adjacent category	Antibodies attack specific cells, as in transfusion reactions.
B	Adjacent category	Complexes deposit in tissues over hours to days.
C	Confusion set	Takes 24 to 72 hours.
D	Correct	IgE triggers mast cells to release histamine within minutes.

6. Answer B Concept 4 • transfer

A rash appears 48 hours after skin contact with poison ivy. Which mechanism is responsible?

Target word: reverse: delayed reaction **Decides:** Delayed onset after contact.

A	Dropped element	Drops the immune sensitization that explains the delay.
B	Correct	Contact dermatitis is delayed and T-cell driven.
C	Adjacent category	Not the mechanism of contact dermatitis.
D	Inverted operation	Type I is immediate, the opposite timing.

7. Answer C Concept 5

A 55-year-old ~~warehouse manager~~ who smokes has crushing chest pain that has lasted 40 minutes and is not relieved by rest. ~~His favorite hobby is fishing.~~ What is the most likely underlying process?

Target word: process **Decides:** Prolonged pain at rest.

A	Dropped element	Drops the thrombus that usually causes sudden blockage.
B	Adjacent category	Inflammation of the heart sac, often with positional pain.
C	Correct	Prolonged ischemia from thrombosis leads to infarction.
D	Confusion set	Stable angina eases with rest in minutes.

8. Answer C Concept 6

A 70-year-old ~~former factory worker~~ with a 40-pack-year smoking history has a barrel chest and trouble breathing out. ~~He lives with two cats.~~ Spirometry shows a reduced FEV1/FVC ratio. What is the main problem in emphysema?

Target word: core defect **Decides:** Air trapping with low FEV1/FVC.

A	Inverted operation	That is restrictive disease.
B	Confusion set	The hallmark of chronic bronchitis.
C	Correct	This defines emphysema as an obstructive disease.
D	Camouflage	Pets do not explain emphysema.

9. Answer A Concept 7

After surgery at a community hospital ~~with a new MRI scanner~~, a patient who received large doses of opioids is breathing slowly. ABG: pH 7.30, PaCO₂ 55 mmHg, HCO₃ 26 mEq/L. ~~The surgery lasted 2 hours.~~ Which imbalance is present?

Target word: acid-base imbalance **Decides:** Low pH with high PaCO₂; normal HCO₃.

A	Correct	Slow breathing retains CO ₂ , lowering pH.
B	Wrong stem number	26 is within normal range (22–26).
C	Confusion set	Would show low HCO ₃ .
D	Inverted operation	Would show high pH and low PaCO ₂ .

10. Answer B Concept 7 · transfer

ABG: pH 7.48, PaCO₂ 40 mmHg, HCO₃ 32 mEq/L. Which imbalance is present?

Target word: reverse: metabolic cause **Decides:** High pH with high HCO₃; normal PaCO₂.

A	Dropped element	Drops the abnormal pH and HCO ₃ .
B	Correct	Excess bicarbonate raises pH.
C	Inverted operation	Would show low pH and low HCO ₃ .
D	Confusion set	Would show low PaCO ₂ .

11. Answer A Concept 8

In a long-time smoker ~~who sings in a choir~~, the normal ciliated columnar cells lining the bronchi are replaced by stratified squamous cells. ~~He also walks every day.~~ What adaptation is this?

Target word: adaptation **Decides:** One mature cell type replaced by another.

A	Correct	Replacement of one mature cell type by another.
B	Confusion set	Disordered, abnormal growth that can come after metaplasia.
C	Adjacent category	More cells of the same type.
D	Adjacent category	Larger cells of the same type.

12. Answer C Concept 8 · transfer

A weightlifter's skeletal muscle cells grow larger without increasing in number. Which adaptation is this?

Target word: reverse: description given, name it **Decides:** Bigger cells, same count.

A	Adjacent category	Cell type changes.
B	Inverted operation	Cells shrink.
C	Correct	Cells enlarge.
D	Confusion set	More cells, not bigger ones.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	11	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	2	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Type 1 vs. type 2 diabetes	Type 1 is autoimmune destruction of pancreatic beta cells (absolute insulin lack); type 2 is insulin resistance with relative deficiency.	Assuming family history of type 2 decides the type.	Islet autoantibodies and very low insulin.
2	Acute inflammation	Vasodilation causes redness and heat; increased vascular permeability lets fluid leak out, causing swelling.	Crediting swelling to vasoconstriction or antibodies.	Redness, heat, swelling, and pain within hours.
3	Left- vs. right-sided heart failure	Left-sided failure backs up into the lungs (crackles, orthopnea); right-sided backs up into the body (JVD, peripheral edema).	Linking edema to left-sided failure or crackles to right-sided.	Lung signs vs. body signs.
4	Hypersensitivity types	Type I: IgE and mast cells, within minutes (anaphylaxis); Type II: antibodies against cells; Type III: immune complexes; Type IV: delayed, T-cell mediated (contact dermatitis).	Calling any allergy Type I.	Minutes vs. 48 hours.
5	Myocardial infarction	MI usually follows rupture of an atherosclerotic plaque with a clot that blocks a coronary artery, causing prolonged ischemia and cell death.	Treating pain unrelieved by rest as stable angina.	Crushing chest pain lasting more than 20 minutes at rest.
6	Emphysema as obstructive disease	Emphysema destroys alveolar walls and elastic recoil, trapping air and lowering FEV1/FVC.	Calling it restrictive or blaming mucus alone.	Barrel chest and trouble exhaling.
7	Acid-base interpretation	pH shows acidosis or alkalosis; a PaCO ₂ moving opposite to pH points to a respiratory cause, and an HCO ₃ moving in the same direction as pH points to a metabolic cause.	Reading a normal HCO ₃ as the cause.	Which value is abnormal in the matching direction.
8	Cellular adaptation	Atrophy = smaller cells; hypertrophy = larger cells; hyperplasia = more cells; metaplasia = one mature cell type replaced by another; dysplasia = disordered, precancerous growth.	Confusing metaplasia with dysplasia, or hypertrophy with hyperplasia.	Cell type replaced vs. cells bigger vs. more cells.

Course scope (public description): Pathophysiology is an overview of the pathology and treatment of diseases in the human body and its systems. This course will explain the processes in the body that result in the signs and symptoms of disease, as well as therapeutic procedures in managing or curing the disease. The content draws on a knowledge of anatomy and physiology to understand how diseases manifest themselves and how they affect the body.